

Model Changelog: v5 to v6

Every change, its evidence, and its academic precedent

****Date:**** 2026-09-16

****Basis:**** v5 production trace, seed 20260914, 4 chains x 2000 draws

****Status:**** every change below is evidence-driven; the evidence is reproducible from the v5 trace

Summary of changes

Change Severity Evidence

- 1 R0 routed into every likelihood via an R0-indexed ODE surrogate CRITICAL v5 raw variables sat exactly at their $N(0,1)$ priors
- 2 Ebola observable corrected from prevalence $I(t)$ to incidence $\sigma E(t)$ CRITICAL constant-beta R0 moves 1.02 $\rightarrow$ 1.92; residual lag-1 ACF 0.678 $\rightarrow$ -0.037
- 3 Influenza observable corrected from $I(t)$ to confinement $Q(t)$ via SEIQR CRITICAL Avilov et al. (2024); v5 ODE peaked day 36 vs data day 6
- 4 Influenza seed offset $t_{\text{seed}} = 11$ days IMPORTANT index case documented 12 days before the series starts
- 5 Ebola GRW removed by default IMPORTANT pre-fit decision rule not met once the observable is correct
- 6 GRW re-parameterised non-centred IMPORTANT centred form produced divergences and ESS ~ 3 once R0 was live
- 7 Ebola GRW (when enabled) indexed by observation with $\sqrt{dt}$ scaling IMPORTANT 53 excluded days make the series irregularly spaced
- 8 $\log_{\text{intercept}}$ priors tightened to $\text{Normal}(0, 0.5)$ IMPORTANT a wide intercept re-absorbs the scale that identifies R0
- 9 Literature R0 offset removed from the non-centred Deterministic MODERATE double-counts literature once R0 is identified
- 10 σ_{rw} justification changed to prior predictive checking MODERATE the v5 "Lemma 1 (Durbin & Koopman sec 2.3)" citation was overstated
- 11 flu_alpha_nb documented as unidentified rather than estimated MODERATE posterior/prior mean ratio 1.27 at HN(50), 1.13 at HN(200)

1. CRITICAL: R0 never entered the likelihood (Ebola, Influenza)

Evidence

v5 solved the ODE once, outside the model, at prior-mean rates:

```
```python
e_base = ode_prevalence(EBOLA_PRIOR.log_beta_mu, ...) # fixed numpy array
e_mu_obs = exp(log_intercept_e + log_rw_e[t]) * e_base[t] # R0 absent
```
```

The sampled `e_R0` and `flu_R0` were `pm.Deterministic` transformations of the hyperprior and a raw variable that appeared nowhere in the likelihood. From the v5 production trace:

```

```
e_log_R0_raw mean = -0.0157 sd = 1.0105 prior N(0,1) NOT UPDATED
flu_log_R0_raw mean = -0.0005 sd = 1.0124 prior N(0,1) NOT UPDATED
n_log_R0_raw mean = -0.6206 sd = 0.8949 prior N(0,1) updated
```
```

Only Norovirus routed R0 into a likelihood, through the Kermack-McKendrick final-size relation. The reported "Ebola R0 = 2.2" and "Influenza R0 = 1.7" were therefore the Norovirus-driven hyperprior rescaled by each arm's literature offset, not estimates from Ebola or Influenza data.

This also explains every anomaly previously attributed to other causes:

- ESS near 7,000 for `e_R0` and `flu_R0` -- sampling an unconstrained prior is easy
- CV of 216% -- that is the hyperprior width, not a posterior width
- `flu_R0` barely moving when `sigma_rw_flu` was halved from 0.30 to 0.15 -- the parameter was never connected to the data in the first place

Fix

`core/bhmm/ode_surrogate.py` precomputes trajectories on a log-uniform R0 grid and interpolates differentiably inside the model:

```

```
idx = (log R0 - log R0_min) / dlog
```

```
i0 = floor(idx); w = idx - i0
traj(R0) = (1-w) * GRID[i0] + w * GRID[i0+1]
```

```

Verified post-fix gradients at the initial point:

```
```
e_log_R0_raw |d logp/d theta| = 27.1 identified
flu_log_R0_raw |d logp/d theta| = 4002 identified
n_log_R0_raw |d logp/d theta| = 542.4 identified
```
```

Precedent

Emulating an expensive simulator on a parameter grid and interpolating is standard in Bayesian calibration of computer models: Kennedy MC & O'Hagan A (2001) J R Stat Soc B 63(3):425-464; Conti S & O'Hagan A (2010) J Stat Plan Inference 140(3):640-651. We use exact ODE solutions rather than a GP emulator, so there is no emulator uncertainty to propagate, only bounded interpolation error, which is measured and written to `tables/ode\_grid\_interpolation\_error.json`.

Permanent guard

`core/diagnostics.identifiability\_gate` computes $d \log p / d \theta$ for every estimated parameter before sampling; `scripts/04\_run\_bhmm.py` aborts if any gradient is zero. `tests/test\_unified\_engine.py::test\_r0\_identifiable` enforces the same condition in CI.

2. CRITICAL: wrong observable for Ebola

`onset` counts NEW symptom onsets per day: an incidence flux. v5 compared it to the prevalence $I(t)$, which integrates and lags onsets by the infectious period. The correct observable is the flux into the symptomatic compartment, $\sigma * E(t)$.

Measured on the eligible rows with a constant-transmission fit:

Observable Fitted R0 Residual lag-1 ACF Ljung-Box p

--- --- ---

prevalence $I(t)$ (v5) 1.02 +0.678 < 1e-10

incidence $\sigma * E(t)$ (v6) 1.92 -0.037 0.885

The v6 value of 1.92 falls inside the Althaus (2014) range of 1.51-2.53 (central 1.83). The v5 value did not.

3. CRITICAL: wrong observable for Influenza

`in\_bed` counts boys CONFINED TO BED. Boys were infectious roughly two days before confinement and remained in bed roughly five days, largely withdrawn from contact. `in_bed` is therefore a confinement compartment downstream of the infectious compartment, not the infectious compartment itself.

```
> "it is erroneous to interpret the 'confined-to-bed' persons (B) as the
> 'infected and infectious' (I) in the classical SEIR model"
> -- Kalachev et al., discussed in Avilov KK, Li Q, Lin L, Demirhan H, Stone L,
> He D (2024). The 1978 English boarding school influenza outbreak: where the
> classic SEIR model fails. J R Soc Interface 21(220):20240394.
```

v6 uses $S \rightarrow E \rightarrow I \rightarrow Q \rightarrow R$ with `in_bed` mapped to Q, and Q excluded from the force of infection.

Symptom of the v5 error: the v5 ODE base trajectory rose monotonically from 1.0 to 6.85 across the 14-day window and peaked at day 36, while the data peaks at day 6 with 282. The GRW was therefore doing 100% of the shape work.

4. IMPORTANT: Influenza seed offset

The index case was infected by 1978-01-10 and febrile 1978-01-15 to 01-18; the series begins 1978-01-22 (RECON `influenza\_england\_1978\_school`; BMJ 1978;1(6112):587). Forcing model time 0 onto observation day 0 assumes the epidemic began when recording began.

t_{seed} is fixed at 11 days: inside the documented 8-12 day window, and the value that minimises the Poisson deviance of the constant-beta fit within it. It is NOT estimated -- with 14 observations, jointly identifying t_{seed} and R_0 reintroduces the non-identifiability this version exists to remove. Sensitivity over {7, 9, 11, 13} is prespecified.

5. IMPORTANT: Ebola GRW removed

The decision rule stated in the project's own GRW documentation is: include a GRW only if the residuals of the constant-transmission fit show $|\text{lag-1 ACF}| > 0.20$ or Ljung-Box $p < 0.05$.

v5 measured the ACF on the RAW data (lag-1 = +0.678) rather than on residuals from a fitted mechanistic model. With the corrected observable, the residual lag-1 ACF is -0.037 with Ljung-Box $p = 0.885$. The rule is not met.

The Ebola GRW was compensating for the wrong observable, not for genuine time-varying transmission. Removing it deletes 139 latent parameters and takes the latent-to-observation ratio from $193/139 = 1.39$ down to 0. This addresses the v5 Ebola R_0 CV of 216% at its cause rather than by tightening a prior.

A GRW can still be forced with `--grw-ebola` for sensitivity.

6. IMPORTANT: non-centred GRW

The centred form puts σ_{rw} in the conditional scale of every latent state, creating a funnel between σ_{rw} and the walk -- the same pathology that the non-centred R_0 reparameterisation removes. It was harmless in v5 only because R_0 was not in the likelihood and the walk had nothing to compete with.

With R_0 live, a 2-chain trial of the centred form gave 4 divergences, Rhat 1.46 and ESS 3 on flu_{R_0} and $\sigma_{\text{rw_flu}}$. The non-centred equivalent:

...

```
eps_t ~ Normal(0, 1)
log_rw_t = sigma_rw * sum_{k<=t} sqrt(dt_k) * eps_k
...
```

gives $\text{Var}[\log_{\text{rw}_0}] = \sigma_{\text{rw}}^2$ (init matches innovation) and $\text{Var}[\log_{\text{rw}_t} - \log_{\text{rw}_{t-1}}] = \sigma_{\text{rw}}^2 * dt_t$, i.e. the same process with Normal(0,1) sampling geometry.

Precedent: Betancourt M & Girolami M (2015), Hamiltonian Monte Carlo for Hierarchical Models; Papaspiliopoulos O, Roberts GO & Skold M (2007) Stat Sci 22(1):59-73.

9. MODERATE: literature R_0 offset removed

v5 added `log( $R_{0\_base\_d}$ )` inside the non-centred Deterministic, shifting each arm's prior location toward its literature value. That was harmless only while R_0 was unidentified. Now that the data informs R_0 , retaining the offset would double-count the literature -- once in the prior location and again when a reader compares the posterior to the same literature range. Each arm's R_0 is now driven by its own likelihood plus the shared hyperprior only.

10. MODERATE: honest prior justification for sigma_rw

v5 justified sigma_rw with

```
---
```

```
sigma_max = log(R0_hi / R0_lo) / sqrt(T)
```

```
---
```

presented as "Lemma 1 (Durbin & Koopman 2012, sec 2.3)" with a proof. **That citation is overstated.** Durbin & Koopman sec 2.3 discusses the signal-to-noise ratio of the local level model; it does not contain that lemma. A reviewer checking the reference would not find it. The formula is a reasonable heuristic but it is the author's own construction.

v6 justifies the prior by prior predictive checking, which is a standard, citable procedure with a defined place in the Bayesian workflow:

- Gelman A et al. (2020). Bayesian Data Analysis, 3e. CRC Press, sec 6.4.
- Gabry J, Simpson D, Vehtari A, Betancourt M & Gelman A (2019). Visualization in Bayesian workflow. J R Stat Soc A 182(2):389-402.
- Schad DJ, Betancourt M & Vasisht S (2021). Toward a principled Bayesian workflow in cognitive science. Psychological Methods 26(1):103-126.

`scripts/03\_prior\_predictive\_calibration.py` simulates the GRW forward from each candidate prior and reports the implied multiplicative range of beta(t) across the observation window. A prior is accepted when the 95th percentile of that range stays below a 3x plausibility ceiling.

Results:

Arm T Selected Median beta range p95 Author heuristic (reported, not cited)

```
-----
```

Ebola 139 HalfNormal(0.02) 1.27x 2.22x 0.0438

Influenza 14 HalfNormal(0.10) 1.32x 2.73x 0.2621

The heuristic is still printed alongside, explicitly labelled as an author-derived rule of thumb, so the two lines of reasoning can be compared.

```
---
```

11. MODERATE: flu_alpha_nb is unidentified, not estimated

The Influenza Var/Mean of 103.6 reflects the epidemic arc across 14 days, not within-day count noise, so the NB log-likelihood is nearly flat in the dispersion parameter (Cameron AC & Trivedi PK (2013), Regression Analysis of Count Data 2e, sec 3.3).

Prior Prior mean Posterior mean Ratio

```
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```

HalfNormal(50) 39.9 50.8 1.27

HalfNormal(200) 159.6 180 1.13

Widening the prior did not identify the parameter, exactly as predicted. v6 keeps HalfNormal(200) as a DISCLOSURE choice and reports the parameter as unidentified (Gelman et al. 2020 BDA3 sec 13.3). `tables/prior\_dominance.csv` records this automatically each run.

```
---
```

Verification

```
```bash
```

```
python tests/test_unified_engine.py # includes the identifiability gate
```

```
python scripts/02_predit_diagnostics.py # regenerates all pre-fit evidence
```

```
python scripts/03_prior_predictive_calibration.py
```

```
```
```

```
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```

12. IMPORTANT: ascertainment intercept conditioned on the observation process

Evidence

A Poisson-deviance profile over (R0, log_intercept) for the Influenza arm shows a clean diagonal ridge -- the two parameters trade off almost exactly because both scale the same trajectory:

```
R0 \ log_intercept -0.60 -0.30 0.00 +0.30 +0.60
```

```
---
2.0 4081 3395 2787 2286 **1928**
2.5 1823 1495 **1370** 1520 2042
3.0 976 **947** 1226 1922 3181
3.5 **863** 1000 1503 2502 4169
4.0 **1050** 1263 1871 3010 4866
```

The per-row minimum walks from +0.6 to -0.6 as R0 rises. This ridge saturated the NUTS tree depth: a quick run took over 20 minutes and still had not finished, versus 157 s after the fix.

Fix

An ascertainment intercept is estimated only where the observation process can actually miss cases.

Arm Observation process Intercept Justification

```
---
Ebola retrospective, incomplete surveillance estimated, Normal(0, 0.5) Khan et al. (1999) describe case finding improving sharply once the international team arrived in May 1995
```

Influenza daily census of every boy confined to bed in a closed 763-boy school fixed at 0 (ascertainment = 1) BMJ 1978;1(6112):587 -- the school counted everyone; there is no under-ascertainment to absorb

This is a data-collection fact, not a tuning choice.

13. IMPORTANT: Influenza GRW disabled by default (saturated latent process)

Evidence

With $T = 14$ a daily GRW has 14 free innovations against 14 observations. The latent process can reproduce any data vector exactly, so R0, sigma_rw and alpha_nb become mutually unidentifiable.

Measured, 3-chain trials, identical seed:

Configuration flu_R0 Rhat flu_R0 ESS sigma_rw_flu Rhat flu_alpha_nb divergences

```
---
GRW on 1.80 3 1.82 prior-dominated 0
GRW off **1.00** **1446** n/a **0.93, data-informed** 0
```

With the GRW off, every parameter in the model reaches $Rhat \leq 1.01$ with $ESS \geq 321$. Notably `flu\_alpha\_nb` stops being prior-dominated: its posterior mean is 0.93 against a prior mean of 160, because there is now a real mechanistic trajectory to measure deviation from rather than a latent process that already fits perfectly.

Interpretation

The residual lag-1 ACF of +0.80 remains. That is a genuine, published misfit for this dataset (Avilov et al. 2024), not something that should be absorbed by 14 free parameters at the cost of making R0 unidentifiable. It is reported in `tables/residual\_serial\_correlation.csv` every run and carried into the debug PDF.

Enable with `--grw-influenza` for sensitivity. Precedent for coarsening rather than saturating a time-varying transmission term: Flaxman et al. (2020) Nature 584:257-261 and Abbott et al. (EpiNow2) both use weekly rather than daily steps for exactly this reason.

Net effect on the default model

```

v5 v6
---
Free parameters 240 10
R0 arms identified 1 of 3 3 of 3
GRW latent variables 207 0
Ebola R0 2.2, CV 216% (prior draw) 1.93, sd 0.034 (Althaus range 1.51-2.53)
flu_alpha_nb prior-dominated data-informed
Quick-run time ~35 min ~4 min

```

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```

Change Severity Evidence

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```

```

14 Measles Hagelloch 1861 added as a fourth arm MAJOR v6 rested its three-way comparison on Influenza, which has a documented
identification conflict and 18x worse precision than the other arms
15 Per-disease ODE grid bounds IMPORTANT measles R0 is 12-18; the shared ceiling of 15 would have clipped and silently flattened the
gradient
16 HEADLINE_ARMS / CAVEATED_ARMS split IMPORTANT prevents flu_R0 being reported as a clean comparator by accident
17 Final-size check skipped for Measles MODERATE AR = 188/188 = 100% makes the K-M relation degenerate; reporting a "conflict" there
would be spurious
18 `--drop-influenza` sensitivity switch MODERATE lets the caveated arm's influence on the hyperprior be measured directly

```

14. Measles Hagelloch 1861 added

Data provenance

Extracted from `hagelloch.df` in the R `surveillance` package via the CRAN source mirror, parsed with the pure-Python `rdata` reader. No values were transcribed, generated or imputed. `scripts/00\_extract\_measles.py --verify` re-derives the files from the source object and compares hashes, so the committed CSVs are auditable. Full citation chain in `data/raw/measles/PROVENANCE.md`.

Verified facts: 188 children (complete village cohort), 12 deaths (CFR 6.38%), rash onsets 1861-11-03 to 1862-01-27, 86-day window, peak 26 on day 34, classes preschool 90 / 1st 30 / 2nd 68.

Pre-fit verdict

Quantity Value Decision

```
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```

```

Var/Mean 10.64 NegBinomial indicated
Residual lag-1 ACF -0.005 GRW NOT warranted (rule:  $|r| > 0.20$ )
Ljung-Box p ~0.99 no residual temporal structure
Constant-transmission R0 8.01 below the classical 12-18; see caveat
Attack rate 188/188 = 100% final-size relation degenerate
Identifiability gradient 221 PASS

```

The arm behaves like Ebola: correct incidence observable, clean residuals, no GRW, ascertainment fixed at 1 because the linelist is a census.

Caveats carried forward

The constant-transmission fit peaks at about 7 against an observed 26, because homogeneous mixing cannot reproduce household and classroom bursts (Neal PJ & Roberts GO (2004) Biostatistics 5(2):249-261 model the contact structure explicitly). NegBinomial dispersion absorbs the spikiness and the residual ACF confirms no trend remains. Guerra FM et al. (2017) Lancet Infect Dis 17(12):e420-e428 find measles R0 estimates spanning 1-770 with median 15.9 and argue the 12-18 canon is over-narrow, so a village estimate below it is not automatically wrong. Compare the posterior 89% ETI, not the point fit.

15. Per-disease ODE grid bounds

```

`DiseaseConfig` now carries `grid_R0_lo` / `grid_R0_hi`. Measles uses
[0.5, 40.0]; the others keep [0.3, 15.0]. A posterior pressed against a grid
bound clips in `ODEGrid.interpolate` and the gradient flattens, which would
re-create the v5 non-identifiability silently.
`tests/test_unified_engine.py::test_measles_grid_covers_literature` enforces the

```

ceiling.

Change Severity Evidence

19 Kermack-McKendrick solver replaced **CRITICAL** 12-step Newton was wrong below $R_0=1$ and corrupted the Norovirus arm
20 Route B added for the undocumented pathogen MAJOR prior-predictive-only arm learns nothing; hierarchical borrowing does
21 Full sampler-process auditing MAJOR the run reported where it ended, not how it got there
22 Influenza reclassified as a BENCHMARK arm IMPORTANT every convergence check passed while the model demonstrably did not fit
23 All figure generation chained automatically IMPORTANT figure scripts had to be invoked by hand

19. CRITICAL: the final-size solver was wrong below $R_0 = 1$

Evidence

The in-model solver used 12 Newton steps from $AR_0 = 1 - \exp(-R_0)$. For $R_0 \leq 1$ the relation $AR = 1 - \exp(-R_0 * AR)$ has the single root $AR = 0$, and Newton approaches a boundary root only LINEARLY. Twelve steps stopped far short:

R_0 12-step Newton true final size

| | | |
|-------|--------|---|
| 0.850 | 0.0933 | 0 |
| 0.900 | 0.1243 | 0 |
| 0.919 | 0.1373 | 0 |
| 1.000 | 0.1983 | 0 |

In the v7 production run the Norovirus posterior settled at **$n_{R_0} = 0.919$** with a reported attack rate of 0.139, close to the observed 0.152. Every convergence diagnostic passed: $\hat{R}$ 1.0019, ESS 2546, zero divergences.

But 0.919 is subcritical. The true final size there is exactly zero. **R_0** was being identified by the solver's truncation error rather than by the Kermack-McKendrick relation. **The correct R_0 for an attack rate of 0.152 is 1.085 .**

`tables/final\_size\_consistency.csv` flagged the contradiction (implied AR 0.000 against reported 0.152) but the model itself did not, and the flag was not treated as blocking.

Fix

Damped fixed-point iteration from $AR = 0.999$, 80 steps, which converges to the true root from above for every R_0 and collapses to ~ 0 when $R_0 \leq 1$:

R_0 fixed-point (80) reference

| | | |
|-------|----------|----------|
| 0.919 | 0.000919 | 0 |
| 1.085 | 0.153282 | 0.152480 |
| 1.500 | 0.582812 | 0.582812 |
| 8.000 | 0.999664 | 0.999664 |

Differentiable throughout, so NUTS gradients still reach R_0 .

Consequence

The v7 Norovirus R_0 of 0.919 must be discarded. Expect approximately 1.085 after the refit. Ebola, Measles and Influenza are unaffected: none of them route R_0 through the final-size relation.

20. Route B for the undocumented pathogen

`core/bhmm/zombie\_arm.py`. Route A (prior predictive) restates the assumption and learns nothing. Route B draws a new arm from the fitted hierarchy, so its range is informed by the measured outbreaks without inventing a single observation.

Measured on the v7 trace:

Route Median 89% interval Width Relative spread

| | | | | |
|---------------------------|------|--------------|------|------|
| A — assumption restated | 3.04 | [2.57, 3.60] | 1.03 | 11% |
| B — learned from the arms | 1.70 | [0.56, 5.08] | 4.52 | 137% |

Route B is 4.4x wider and its median sits between the measured arms. That width is honest uncertainty about an unmeasured pathogen; Route A's narrowness is borrowed confidence from a choice.

Reference: Gelman A et al. (2020) BDA3 sec 5.5 (prediction for a new group).

21. Sampler-process auditing

`core/mcmc\_audit.py` plus `scripts/15\_mcmc\_audit\_report.py` write twelve CSVs covering step size, tree depth, energy and E-BFMI, acceptance rate, log-posterior trace, mass matrix, ESS growth, MCSE decay, rank uniformity, posterior correlation and a per-parameter convergence ledger.

Verified on the v7 trace: E-BFMI 0.82-0.88 across four chains, tree-depth saturation 0.05%, mean acceptance 0.945 against a target of 0.95, zero rank-uniformity failures, strongest posterior correlation $n_{R0} \times n_{\text{attack_rate}}$ at 0.996 (expected — the attack rate is a deterministic function of $R0$).

The log-posterior trace is the closest analogue to a training-loss curve and ESS growth to a learning curve.